2. Random Variables

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Outline

1. Introduction
2. Discrete Random Variables
3. Continuous Random Variables
4. Cumulative Distribution Functions
5. Great Expectations
6. LOTUS, Moments, and Variance
7. Approximations to $\mathrm{E}[h(X)]$ and $\mathrm{Var}(h(X))$
8. Moment Generating Functions
9. Some Probability Inequalities
10. Functions of a Random Variable
11. Inverse Transform Theorem
12. Honors Bonus Results

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Thus, you can speak of $P(X = x)$.

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In addition,

$$P(X = x) = \begin{cases} 1/36 & \text{if } x = 2 \\ 2/36 & \text{if } x = 3 \\ \vdots & \\ 6/36 & \text{if } x = 7 \\ \vdots & \\ 1/36 & \text{if } x = 12 \\ 0 & \text{otherwise.} \end{cases} \quad \square$$

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For our purposes, X and Y are the same, since $P(X=0) = P(Y=0) = \frac{1}{2}$ and $P(X=1) = P(Y=1) = \frac{1}{2}$.

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But $P(X \leq 0.65) = 0.65$ and $P(X \in [0.3, 0.7]) = 0.4$.

If A is any *interval* in [0,1], then $P(X \in A)$ is the length of A.

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Example: The amount of time you wait in a line is either 0 (with positive probability) or some positive real number — a *combined* discrete-continuous random variable!

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Example: A discrete RV can have any values. For instance, let X denote the possible profits from an inventory policy, where $f(-5.1) = 0.2$ (lose money), $f(1.3) = 0.5$ (break even), and $f(11) = 0.3$ (big bucks!).

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Let $X =$ number of 6's. Then $X \sim \text{Bin}(3, \frac{1}{6})$. $\square$

Theorem: If $X \sim \text{Bin}(n, p)$, then the probability of k successes in n trials is

$$P(X = k) \;=\; \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, \ldots, n,$$

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The number of ways to arrange the sequence is $\binom{n}{k}$. Done. $\square$

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$$P(X = 2) \;=\; \binom{n}{k} p^k q^{n-k} \;=\; \binom{3}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^1 \;=\; \frac{15}{216}.$$

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k	0	1	2	3
$P(X = k)$	$\frac{125}{216}$	$\frac{75}{216}$	$\frac{15}{216}$	$\frac{1}{216}$

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So $X \sim \text{Bin}(12, 2/9)$, and then

$$P(X = 3) \;=\; \binom{12}{3}\left(\frac{2}{9}\right)^3\left(\frac{7}{9}\right)^9. \quad \square$$

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$$\begin{aligned} P(X \geq 4) &= 1 - P(X = 0, 1, 2, 3) \\ &= 1 - e^{-10}\left(\frac{10^0}{0!} + \frac{10^1}{1!} + \frac{10^2}{2!} + \frac{10^3}{3!}\right) \\ &= 0.9897. \quad \square \end{aligned}$$

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- Instead, think of $f(x)\,dx \doteq P(x < X < x + dx)$.

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$$P(-1 < X < 6) \;=\; \int_{-1}^{6} \frac{1}{8-(-2)}\,dx \;=\; 0.7. \quad \square$$

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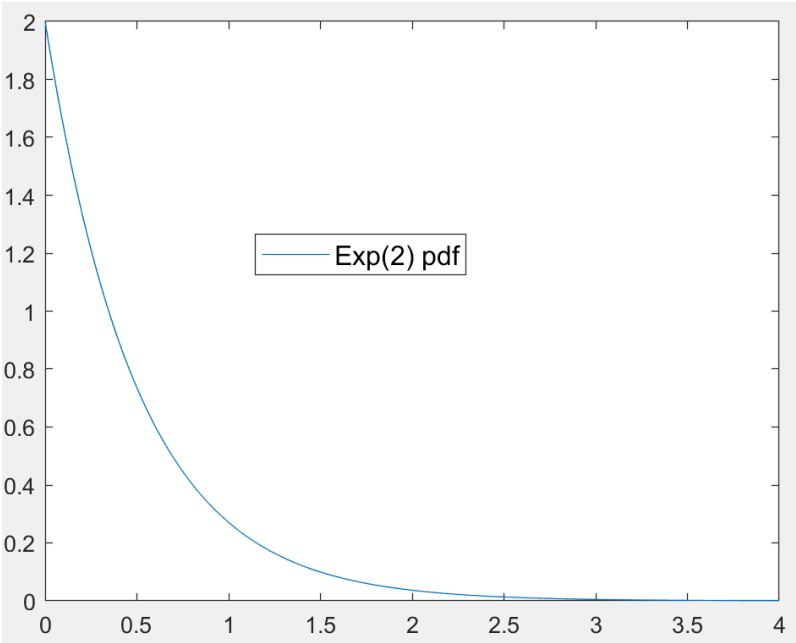

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Now we can calculate any reasonable probabilities, e.g.,

$$P(0 < X < 1) \;=\; \int_0^1 \frac{3}{8}x^2\,dx \;=\; 1/8.$$

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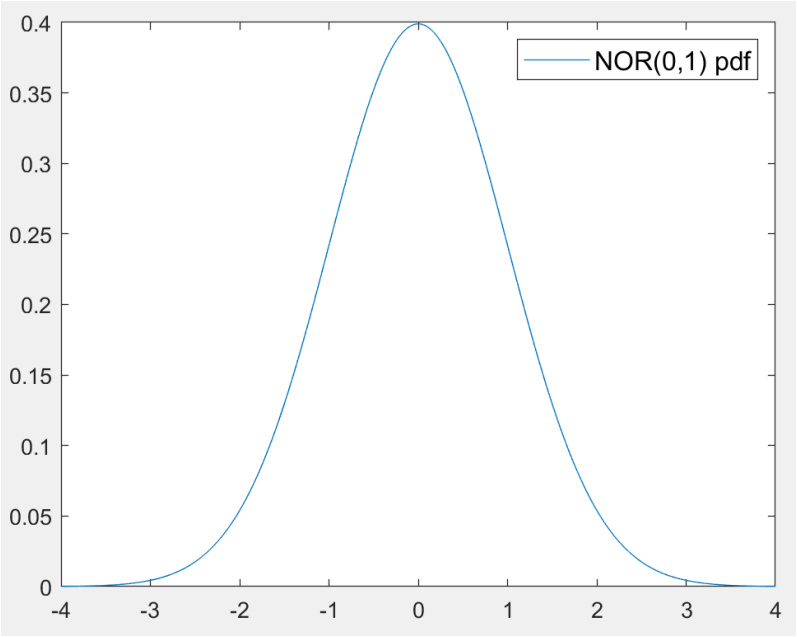

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Lesson 2.4 — Cumulative Distribution Functions

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Warning! For *discrete* RVs, you have to be careful about "$\le$" vs. "$<$" at the endpoints of the intervals (where the step function jumps).

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Solving, we obtain $m = (1/\lambda)\ell\text{n}(2)$. $\square$

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Proof: By complements, $P(X > x) = 1 - P(X \leq x) = 1 - F(x)$. $\square$

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Lesson 2.5 — Great Expectations

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The Next Few Lessons:

- Mean (Expected Value)
- Law of the Unconscious Statistician
- Variance
- Probability Inequalities
- Approximations

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The mean gives an indication of a RV's *central tendency*. Think of it as a weighted average of the possible x's, where the weights are given by $f(x)$.

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So it'll take, on average, $\text{E}[X] = 1/p = 1/0.4 = 2.5$ shots. $\square$

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Outline

1. Introduction
2. Discrete Random Variables
3. Continuous Random Variables
4. Cumulative Distribution Functions
5. Great Expectations
6. LOTUS, Moments, and Variance
7. Approximations to $\mathrm{E}[h(X)]$ and $\mathrm{Var}(h(X))$
8. Moment Generating Functions
9. Some Probability Inequalities
10. Functions of a Random Variable
11. Inverse Transform Theorem
12. Honors Bonus Results

Lesson 2.6 — LOTUS, Moments, and Variance

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Law of the Unconscious Statistician (LOTUS)

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Theorem: The expected value of a function of X, say $h(X)$, is

$$\boxed{\mathrm{E}[h(X)] \equiv \begin{cases} \sum_x h(x) f(x) & \text{if } X \text{ is discrete} \\ \int_{\mathbb{R}} h(x) f(x)\, dx & \text{if } X \text{ is continuous.} \end{cases}}$$

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Examples: $\mathrm{E}[X^2] = \int_{\mathbb{R}} x^2 f(x)\, dx, \quad \mathrm{E}[\sin X] = \int_{\mathbb{R}} (\sin x) f(x)\, dx.$

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The next results establish the fact that the expected value operator can pass through certain linear functions of X, and then can be used to obtain pleasant expressions for other expected values and variances.

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Theorem (easier way to calculate variance):

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Outline

1. Introduction
2. Discrete Random Variables
3. Continuous Random Variables
4. Cumulative Distribution Functions
5. Great Expectations
6. LOTUS, Moments, and Variance
7. Approximations to $\mathrm{E}[h(X)]$ and $\mathrm{Var}(h(X))$
8. Moment Generating Functions
9. Some Probability Inequalities
10. Functions of a Random Variable
11. Inverse Transform Theorem
12. Honors Bonus Results

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both of which are reasonably close to their true values (0.8 and 0.0267, respectively). $\square$

Outline

1. Introduction
2. Discrete Random Variables
3. Continuous Random Variables
4. Cumulative Distribution Functions
5. Great Expectations
6. LOTUS, Moments, and Variance
7. Approximations to $\mathrm{E}[h(X)]$ and $\mathrm{Var}(h(X))$
8. Moment Generating Functions
9. Some Probability Inequalities
10. Functions of a Random Variable
11. Inverse Transform Theorem
12. Honors Bonus Results

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Thus, you can *generate* the moments of X from the mgf. (Sometimes, it's easier to get moments this way than directly.)

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Same deal for higher-order moments. $\square$

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Example: Let $X \sim \mathrm{Exp}(\lambda)$ and $Y = 3X + 2$. Then

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Theorem (mgf of a linear function of X): Suppose X has mgf $M_X(t)$ and let $Y = aX + b$. Then

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Then by a previous example and the uniqueness of mgf's, it *must* be the case that $Y \sim 3X + 2$, where $X \sim \text{Exp}(\lambda)$. $\square$

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Theorem (Chebychev's Inequality):[1] Suppose $\mathrm{E}[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2$.

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Theorem (Chebychev's Inequality):[1] Suppose $\mathrm{E}[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2$. Then, for any $c > 0$,

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Chebychev gives a bound on the probability that X deviates from the mean by more than a constant, in terms of the constant and the variance. You can always use Chebychev, but it's crude.

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In particular, for $c = 1/3$,

$$P\left(\left|X - \frac{1}{2}\right| \geq \frac{1}{3}\right) \leq \frac{3}{4} \quad \text{(upper bound).}$$

Example (cont'd): Let's compare the above bound to the *exact* answer.

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So Chebychev bound of 3/4 was pretty high by comparison. □

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Lesson 2.10 — Functions of a Random Variable

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The Next Few Lessons:

- Problem Statement with Examples
  - Discrete Case
  - Continuous Case
- Inverse Transform Theorem
- Some Honors Topics

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We'll start with the case in which X is a discrete RV, and then we'll go to the continuous X case.

Discrete Case: X discrete implies Y discrete implies

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Example: $f(x) = |x|, -1 \leq x \leq 1.$

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This means that Y has the Unif(0,1) distribution! $\square$

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We can generalize this result. . . .

Outline

1. Introduction
2. Discrete Random Variables
3. Continuous Random Variables
4. Cumulative Distribution Functions
5. Great Expectations
6. LOTUS, Moments, and Variance
7. Approximations to $\mathrm{E}[h(X)]$ and $\mathrm{Var}(h(X))$
8. Moment Generating Functions
9. Some Probability Inequalities
10. Functions of a Random Variable
11. Inverse Transform Theorem
12. Honors Bonus Results

Lesson 2.11 — Inverse Transform Theorem

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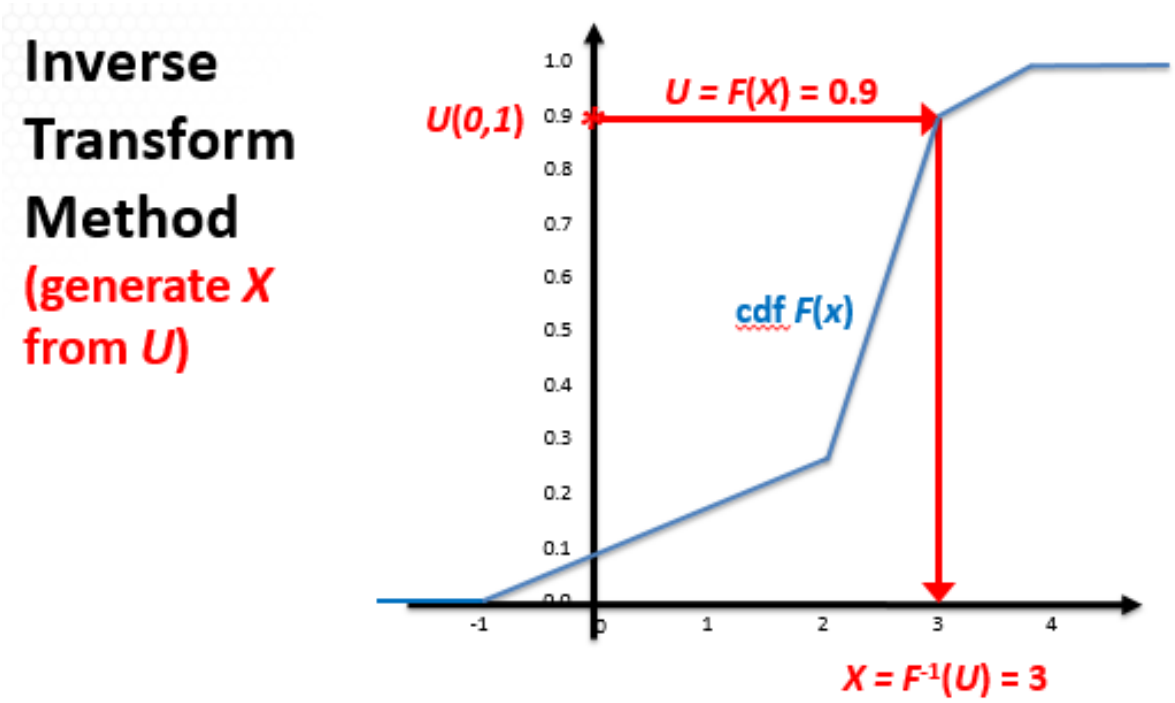

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- Generate 10000 Unif(0,1)'s (e.g., use the `rand` function in Excel, or `unifrnd` in Matlab),

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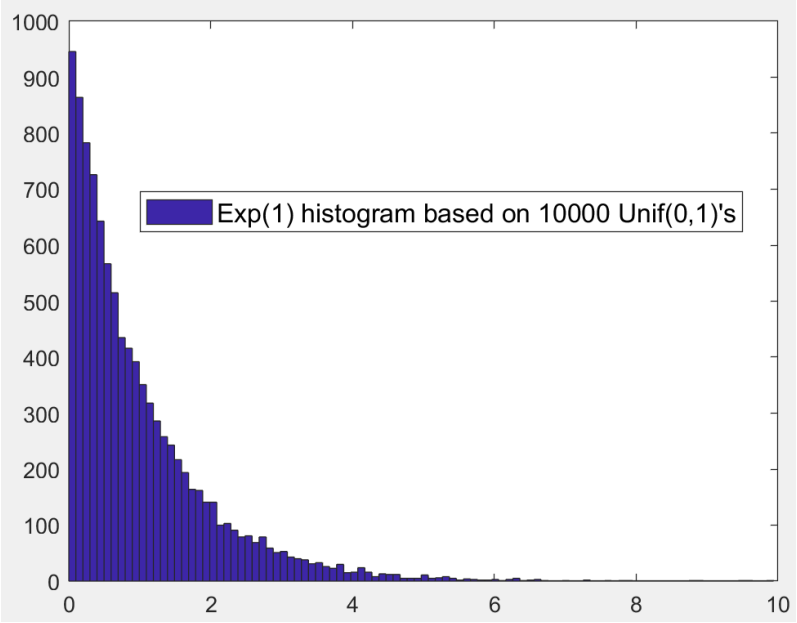
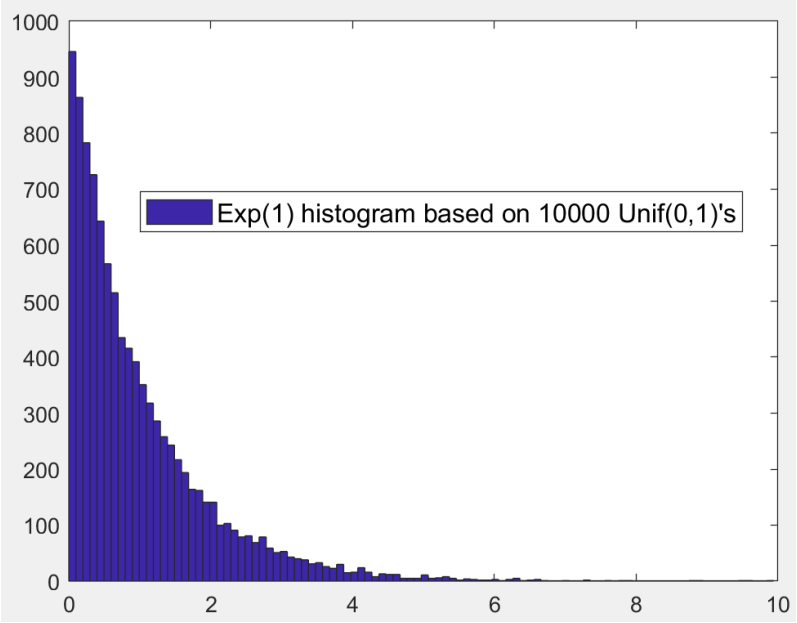

1000
900
800
700
600
500
400
300
200
100
0
0
2
4
6
8
10
Exp(1) histogram based on 10000 Unif(0,1)'s



Remark: This trick has tremendous applications in simulation, where we need to generate random variables all the time (e.g., customer arrival times, service times, machine breakdown times, etc.).

Outline

1. Introduction
2. Discrete Random Variables
3. Continuous Random Variables
4. Cumulative Distribution Functions
5. Great Expectations
6. LOTUS, Moments, and Variance
7. Approximations to $\mathrm{E}[h(X)]$ and $\mathrm{Var}(h(X))$
8. Moment Generating Functions
9. Some Probability Inequalities
10. Functions of a Random Variable
11. Inverse Transform Theorem
12. Honors Bonus Results

Lesson 2.12 — Honors Bonus Results

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Another Way to Find the pdf of a Function of a Continuous RV

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&= \frac{d}{dy}F(h^{-1}(y)) \\
&= f(h^{-1}(y))\left|\frac{dh^{-1}(y)}{dy}\right| \quad \text{(chain rule).} \quad \square
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Next up: Bivariate generalizations of the material in this module!